
Research Statement – Sean Patrick MacBride

My research focuses on exploring the cosmic frontier using astronomical instrumentation, multi-messenger astrophysics, and standard siren cosmology. In this research proposal, I will describe how my research experience and leadership has prepared me for a **Add Position** at **Add Institution**.

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Early Career Contributions

- Commissioning the focal plane array of LSST-Camera, testing and integrating multi-object and integral-field spectroscopic instruments.
- Leading searches for electromagnetic counterparts to gravitational wave mergers using Vera C. Rubin Observatory, and observing campaigns using other photometric and spectroscopic instruments.
- Measuring H_0 using the dark siren method and the Dark Energy Legacy Volume Explorer galaxy catalog.

Astronomical Instrumentation Development: My recent work has spanned both photometric and spectroscopic instruments. This began in 2020, with the **Large Lenslet Array Magellan Spectrograph** (LLAMAS), where I validated the optical performance of the diffraction gratings and collimators for the LLAMAS spectrographs, assembled and verified optical tolerance for the 24 spectrograph cameras, and integrated all 2,400 AR-coated fibers into the focal plane with zero fiber loss, before it was commissioned to the Magellan Baade telescope in 2024 (1; 2). After integrating and testing LLAMAS, I shifted my focus from integral-field spectrographs to multi-object spectrographs. Starting in 2023, while at the University of Michigan-Ann Arbor, I characterized and resolved a common failure mode on Dark Energy Spectroscopic Instrument (DESI) robotic positioners, where I found that reorienting the robotic positioner to an inverted physical configuration could eliminate the failure mode. This indicated that gravity-sag has a significant influence on the presence and recovery of these failures (3). This expertise with spectroscopic instruments translated effectively to the transcendent photometric imager of the next decade, the **LSST Camera** (LSSTCam), where I have served as an LSSTCam Summit Support Scientist since 2024. I led LSSTCam focal-plane commissioning on Cerro Pachón as a principal component of the in-kind contribution from Switzerland to the Rubin Project, testing the CCDs in both the summit clean-room and on the Simonyi Survey Telescope, optimizing the sensor performance, and playing a leading role in achieving LSSTCam first photon in April 2024 (see Figure 1a) (4). **Target-of-Opportunity Searches:** Since 2023, I have led twenty distinct observing campaigns using different telescopes and facilities, with the goal of detecting electromagnetic (EM) counterparts to gravitational wave (GW) merger events. A variety of multi-messenger analyses are enabled by the detection of an EM counterpart to a GW merger event. The *bright siren* method is my particular focus, which uses the redshift information of the host galaxy of an EM counterpart and the luminosity distance inferred from the GW source to constrain cosmology (5). Bright siren searches have been my primary focus while I have have been co-leading the Dark Energy Survey Gravitational Wave follow-up program (DESGW) program since 2023, conducting **Dark Energy Camera** (DECam) observations, processing raw image data from sixteen Target-of-Opportunity (ToO) campaigns to calibrated products, and analyzing processed data in response to LIGO-Virgo-Kagra (LVK) Observing Run 4 (O4) triggers. I led a collaboration between DESGW and the Japanese collaboration for gravitational-wave electromagnetic follow-up (J-GEM) during the O4 observing run when a well-localized gravitational wave merger candidate was accessible to both Northern and Southern hemisphere instruments. In this collaboration, I obtained multi-epoch photometry using DECam, processed the obtained data, and cross-matched this data with spectroscopic observations using **Prime Focus Spectrograph** (PFS), a recently commissioned multi-object spectrograph at the Subaru Telescope. This analysis enabled the first public scientific result using PFS (6). **I have been leading the Vera C. Rubin Observatory (Rubin) ToO observing program as the Rubin ToO observer since 2025.** In this role, I have led observing responses to one interstellar

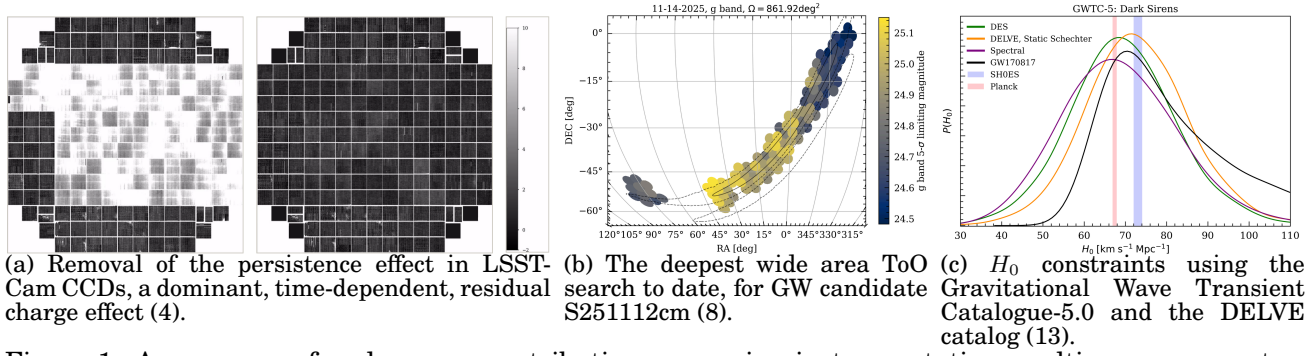


Figure 1: A summary of early career contributions, spanning instrumentation, multi-messenger astrophysics, and standard siren cosmology.

object (7), one high-energy neutrino event (later reclassified as terrestrial), and two gravitational-wave merger candidates, including the deepest, wide-area GW ToO search to date ($m_{g,5\sigma} \sim 24.8$ across $\sim 850 \text{ deg}^2$, see Figure 1b) (8; 9).

Dark Siren Constraints on the Hubble Constant: While recent observing campaigns have not yet yielded an EM counterpart to a GW merger, I have retained a focus on constraining cosmology using EM and GW data, employing the *dark siren* method to measure the expansion rate of the local universe, H_0 . The dark sirens method infers redshift information of GW events using a catalog of potential host galaxies, which is lower precision than the bright siren method, but can be applied to all GW events (10). My work in dark sirens began by supporting the preparation of Dark Energy Survey (DES) data, resulting in the first dark siren constraint using a deep photometric catalog to jointly infer GW population parameters and cosmological parameters (11). DES was accepted as the fiducial result for the most recent LVK analysis, and demonstrated a 22% improvement in constraining H_0 compared to previous analyses (12). I am now expanding on the DES preparation, leading a dark-siren analysis using Dark Energy Local Volume Exploration (DELVE), where initial results show DELVE outperforms DES for dark sirens, and obtains a 12% constraint on H_0 when combined with the bright siren GW170817 (see Figure 1c) (13).

Future Research Activities

- Develop future astronomical instruments, focusing on a complement to Rubin, and deploy monitoring tools for LSST-Camera.
- Lead Target-of-Opportunity searches using Vera C. Rubin Observatory to detect the next bright siren, and other exotic astrophysical events.
- Usher dark siren cosmology into a precision era using LSST data and specialized observational programs.

Astronomical Instruments to Maximize LSST data: In the Legacy Survey of Space and Time (LSST) era, the central bottleneck will not be imaging depth, but rather spectroscopic capacity. To aid in this effort, I will test and deploy advanced focal plane technology for future multi-object spectrographs. As part of this focus, I will explore instrument design for a multi-object spectrograph optimized for standard siren cosmology, specialized for the precision and event rates of forthcoming gravitational wave detectors. This instrument would serve two goals: 1) characterize astrophysical transients associated with potential bright standard sirens, and 2) obtain spectroscopic redshifts from host galaxies to bolster dark siren measurements. Through this dual purpose, the proposed instrument would become a dark energy science machine primed to capitalize on the deluge of data from LSST and future GW observatories. These pathfinding activities would help to inform the proposed **Stage-5 Spectroscopic Experiment** (Spec-S5), a future, large-scale cosmic frontier experiment (14). Developing and testing of focal-plane technology will help shape the capabilities of Spec-S5, so it serves as an effective spectroscopic complement to Rubin starting in the mid-2030's.

I will continue in a support-scientist role for LSSTCam as the LSST transitions into survey operations, maintaining the operational knowledge needed to respond quickly to sensor anomalies and focal-plane issues as they arise. Building on my commissioning-era characterization of focal-plane defects, I will deploy a defect-monitoring tool, currently in-development, to `cp_verify`, the calibration-product verification framework used by the LSST Science Pipelines (15). This tool will elevate defect monitoring to an automated process, directly improving the quality of LSST data.

Bright Siren Searches Enabled by Rubin Observatory: I will lead the observing planning, execution, and analysis of the next bright standard siren accessible to Rubin,

along with other ToO's pursued during the LSST, continuing my role as the Rubin ToO observer throughout the transition to operations. While it is difficult to predict the timing of the next bright siren, I will continue to develop and deploy improvements to the ToO system to improve response latency, observing strategy, and system automation to maximize the forthcoming opportunity to detect and characterize these optical counterparts to obtain a well-constrained measure of H_0 .

In parallel to operating the Rubin ToO system, I will support the analysis of DESGW candidates identified during LVK O4, and lead the analysis of DESGW candidates identified during LVK intermediate runs. This work positions me to sustain and support Rubin's ToO program with additional imaging in other filters or to extended time-frames using a similarly capable photometric imager. This increased temporal sampling and color characterization will play an important role, especially as other priorities of the LSST are balanced against the ToO program. As LSST's ToO program matures, I will work to enhance counterpart characterization by securing spectroscopic follow-up time on other instruments, moving beyond photometric classification to confirm spectral features that can inform the scope of Rubin's ToOs and constrain the physical properties of the underlying transients and their host environments.

Dark Siren Measurements for Precision Cosmology: I will pursue a cutting-edge program that produces a gold-standard galaxy catalog for dark siren cosmology using LSST data. This program proceeds in four stages, each building towards a powerful, world leading measurement. It begins with an ongoing project, focused on characterizing the LSST Data Preview 2 (DP2) galaxy catalog, focusing on robust object classification that is scalable to future LSST data releases (16). Following characterization of DP2, I will finish a forecasting study that quantifies the H_0 precision achievable by combining simulated first-year LSST data with an O4-like gravitational-wave catalog, extending the dark-siren methodology from my recent DELVE measurement to project the constraining power in the LSST era. Superceding DP2 preparation, I will characterize the first LSST Data Release (DR1) galaxy catalog and produce a dark siren measurement of H_0 . This result will extend the methodology I developed with DELVE to a dataset with substantially greater depth, area, and photometric uniformity, all features that will make it the standard galaxy catalog for dark siren analyses. Building on the DR1 measurement, I aim to combine LSST large-scale structure measurements in a joint cosmological analysis, using cosmic shear and standard sirens to obtain a tighter constraint of H_0 , and extend the constraints to Ω_m .

As the wealth of GW data increases alongside a wide variety of new galaxy catalog data, I will explore the frontier of the standard siren method using specialized surveys and forthcoming data releases from other surveys, while engaging with the Horizon Europe-funded GWCats collaboration, a tight group of researchers focused on using galaxy catalogs and GW data to constrain cosmology. To assess the frontier of the dark siren method using observations, I will lead a dedicated spectroscopic dark-siren pilot survey, obtaining targeted redshifts within GW localization

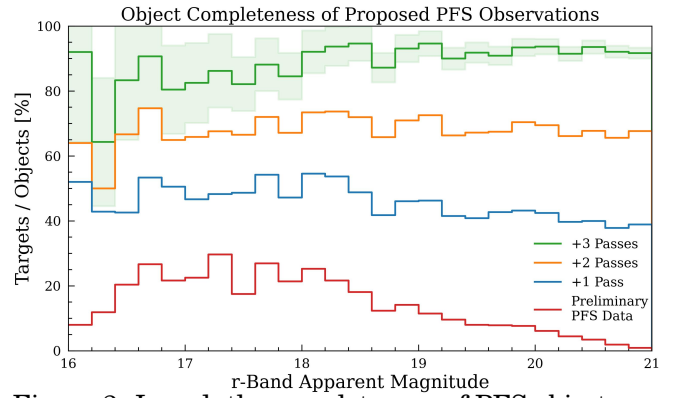


Figure 2: In red, the completeness of PFS objects acquired as part of our ToO campaign (6), with other colors showing the forecasted completeness of proposed observations. The high completeness obtained over the localization region in 3 passes (one night) demonstrates the utility of current and future multi-object spectrographs as characterization machines for standard siren cosmology.

volumes to assess the limits and enhancements to H_0 constraints using well-localized dark sirens. This survey is led through a continuing collaboration with J-GEM on the PFS instrument, obtaining additional observations for the GW merger pursued in our collaborative DESGW ToO with an expected observation date starting in April 2027 (see Figure 2).

Working directly with **Specify specific people to work with** at **Add Institution** will help to develop **Specify targeted project that works very well with**, along with contributing to the research goals and mission of **Add Institution** by **Reference the specific mission of the research institute**. Through this concerted effort, I will drive advancements in astronomical instrumentation, push the boundaries of multi-messenger astrophysics, and mature standard sirens, both bright and dark, into a precision probe in the LSST era.

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